Logistics Performance Analysis

Dataset: E-Commerce Shipping Data | 10,999 records

Key Performance Indicators

Total Shipments	10,999
Overall Delay Rate	59.7%
On-Time Rate	40.3%
Avg Customer Rating	2.99 / 5.0

Executive Summary

Analysis of 10,999 shipments reveals a systemic delivery performance issue affecting 59.7% of all orders. The problem is not isolated to a specific warehouse or shipment mode - all operational segments perform similarly. Two critical factors were identified as primary drivers of delays: discount policy (orders with discounts above 10% show 100% delay rate) and product weight range (2.5-4kg products show 99.9% delay rate). Immediate action on these two levers could significantly improve on-time delivery performance.

Operations Manager Perspective

Finding 1 - Warehouse Performance

All warehouse blocks (A-F) show similar delay rates between 58.6% and 60.2%. No single warehouse is responsible for the overall delay problem.

Finding 2 - Shipment Mode

Road (58.8%), Ship (59.8%), and Flight (60.2%) perform almost identically. Mode of shipment does not explain delays.

Finding 3 - Weight Impact

Products in the 2.5-4kg range show a 99.9% delay rate vs 43.1% for Heavy (4-6kg). The operation is optimized for high-volume weight ranges only.

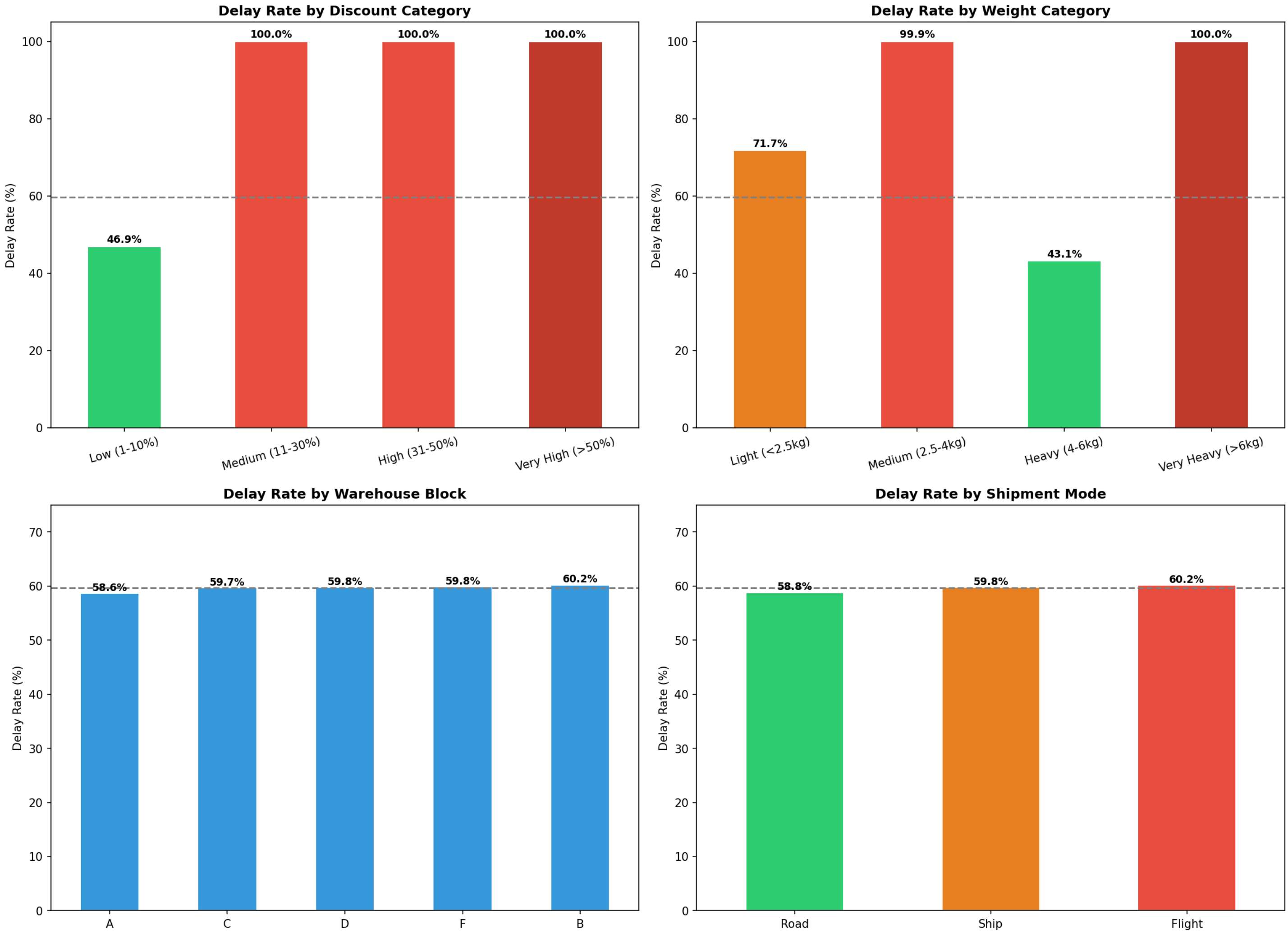
Finding 4 - Discount Policy

Orders with discounts above 10% show 100% delay rate. Discount policy is the strongest predictor of delivery failure identified in this analysis.

Operations Dashboard

E-Commerce Logistics - Executive Report

E-Commerce Logistics — Executive Dashboard



Commercial Director Perspective

Finding 5 - Customer Calls vs Satisfaction

Average customer rating is 3.0 across all call volume segments. More calls do not improve satisfaction, but customers who call more experience fewer delays - suggesting informal prioritization.

Finding 6 - Product Importance vs Delivery

High importance products show the worst delay rate (64.98%). The importance flag is not being used operationally to prioritize shipments.

Finding 7 - Customer Loyalty vs Service

Frequent buyers (7+ purchases) experience 62.67% delay rate - nearly identical to new customers (63.46%). No loyalty-based service differentiation exists.

Finding 8 - Warehouse Ranking

Warehouse A ranks best overall (58.65% delay rate) while B ranks worst (60.23%). However, the difference is marginal and not statistically significant.

Strategic Recommendations

IMMEDIATE	1. Review discount policy - cap discounts at 10% or create operational capacity before offering higher discounts. Current policy guarantees delivery failure above this threshold.
SHORT TERM	2. Investigate 2.5-4kg weight range - identify packaging, handling, or routing issues specific to this segment. This range shows near-total delivery failure.
SHORT TERM	3. Integrate product importance flag into dispatch workflow - high importance products must trigger operational priority, not just commercial priority.
MEDIUM TERM	4. Implement customer loyalty service tiers - frequent buyers should receive measurable service advantages to reduce churn risk.
MEDIUM TERM	5. Deploy formal prioritization system - replace informal call-based prioritization with a rule-based system that protects all customers equally.

Report generated automatically with Python | Project: ecommerce-logistics-bi